



General Features

Sensor	High sensitivity geophone (5Hz/10Hz) KCK Socket option for external geophone string
Digitalization	32-bit Δ - Σ ADC
Storage Capacity	16GB (45 days continuous recording @1ms) 32GB for option
Battery Life	Internal rechargeable Li-Ion battery(69WH) More than 720 hours continuous recording
Timing Accuracy	+/- 10 μ s(continuous GNSS disciplined) +/- 1ms (within 1 hour without GNSS signal)
Automated Tests	Onsite full signal chain function and performance self-inspection and diagnosis
Working Mode	Autonomy with BLE wireless on-site QC inspection; Support UAV patrol
LED Status Indicator	Nodal health, GNSS synchronization state, Wireless data harvesting status
Data Harvest	Infield: Wireless/Portable download unit Indoor: Data harvest/battery charging rack
Infield Management	Industrial handheld tablet
Weight	0.9kg
Size	10.7cm x 10.7cm x 12.7cm
Operating Temperature	-40°C ~ +70°C



Product Highlights

- Customized with Sinopec Geophysical Corporation;
- All-in-one seismic node with build-in 5Hz/10Hz high sensitivity geophone, support external string with KCK adapter;
- 32-bit Δ - Σ AD converter for high resolution, high dynamic range seismic signal recording;
- High sensitive GNSS module equipped with active high-gain patch antenna supports deep planting applications (typically 1m buried depth in sand) ;
- Built-in high-precision signal generator supports on-site full signal chain function and performance self-inspection and diagnosis, including acquisition system noise, real-time dynamic range, geophone impedance, natural frequency, sensitivity, damping, distortion, etc.;
- P&R (Place & Record) mode for convenient infield deployment;
- BLE wireless module together with visual LEDs support onsite configuration and proximity;
- Onsite working status verification after deployment, including battery voltage, environmental temperature, station tilting, environmental signal strength, real-time seismic waveform inspection, etc.;
- High strength engineering plastic enclosure for enhanced waterproof and anti-dropping capability;
- Small volume and light weight for convenient field transportation and deployment.



Acquisition Specifications

ADC Resolution	32-bit	
Sampling Interval	0.25ms, 0.5ms, 1ms, 2ms, 4ms	
Preamplifier Gain	x1, x2, x4, x8, x16, x32, x64 (0dB, 6dB, 12dB, 18dB, 24dB, 30dB, 36dB)	
Input Impedance	40K Ω 22nF (differential mode)	
Gain Accuracy	0.1%	
Full Scale Input Signal	± 2.5 Vp-p@x1gain ± 156 mVp-p@x16gain	± 625 mVp-p@x4gain ± 39 mVp-p@x64gain
Instantaneous DR@4ms	128dB @ x1 gain 120dB @ x16 gain	126dB @ x4 gain 112dB @ x64 gain
Full Scale DR	148dB	
Equivalent Input Noise@4ms	0.7 μ V @ x1 gain 0.08 μ V @ x16 gain	0.2 μ V @ x4 gain 0.06 μ V @ x64 gain
Harmonic Distortion	-120dB @ 31.5Hz	
Frequency Response@0.25ms	0Hz ~ 1652Hz	
Anti-alias Filter	Linear or minimum phase filter, 82.6% of Nyquist -3dB bandwidth	
DC-blocking Filter	0.1 ~ 10Hz	
Stopband Attenuation	>130dB @Nyquist	
Common Mode Rejection	>110dB	